Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim:** Use appropriate smoothing and sharpening techniques to enhance the clarity of the images.	
Open Ended Problem	**Date:**	**Enrollment No: 92510133002**

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os

image_paths = [
    r"1.png",
    r"2.png",
    r"3.png",
    r"4.png",
    r"5.png"
]

image_names = [
    "Image 1",
    "Image 2",
    "Image 3",
    "Image 4",
    "Image 5"
]

output_folder = r"C:\Users\LADVA\Desktop\SEM 5\DSIP\OPE 9"
os.makedirs(output_folder, exist_ok=True)

for i, image_path in enumerate(image_paths):

    if not os.path.exists(image_path):
        continue

    img = cv2.imread(image_path, cv2.IMREAD_UNCHANGED)

    if img is None:
        continue

    if len(img.shape) == 3 and img.shape[2] == 4:
        b, g, r, alpha = cv2.split(img)

        background = np.ones_like(img[:, :, :3]) * 255
        bgr = cv2.merge([b, g, r])

        alpha_float = alpha.astype(float) / 255.0
        alpha_float = alpha_float[:, :, np.newaxis]
```

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```
    img = (
        bgr.astype(float) * alpha_float
        + background.astype(float) * (1 - alpha_float)
    ).astype(np.uint8)

if len(img.shape) == 3:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    original_display = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
else:
    gray = img.copy()
    original_display = img.copy()

mean_filter = cv2.blur(gray, (5, 5))

gaussian_filter = cv2.GaussianBlur(gray, (5, 5), 0)

median_filter = cv2.medianBlur(gray, 5)

bilateral_filter = cv2.bilateralFilter(
    gray,
    9,
    75,
    75
)

laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian = np.absolute(laplacian)

laplacian = cv2.normalize(
    laplacian,
    None,
    0,
    255,
    cv2.NORM_MINMAX
).astype(np.uint8)

laplacian_sharp = cv2.addWeighted(
    gray,
    1.0,
    laplacian,
    1.0,
    0
)
```

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```
    blur_for_unsharp = cv2.GaussianBlur(gray, (5, 5), 0)

    unsharp = cv2.addWeighted(
        gray,
        1.5,
        blur_for_unsharp,
        -0.5,
        0
    )

    blur_for_highboost = cv2.GaussianBlur(gray, (1, 1), 0)

    high_boost = cv2.addWeighted(
        gray,
        2.0,
        blur_for_highboost,
        -1.0,
        0
    )

    number = i + 1

    cv2.imwrite(
        os.path.join(output_folder, f"image_{number}_original.png"),
        gray
    )

    cv2.imwrite(
        os.path.join(output_folder, f"image_{number}_mean.png"),
        mean_filter
    )

    cv2.imwrite(
        os.path.join(output_folder, f"image_{number}_gaussian.png"),
        gaussian_filter
    )

    cv2.imwrite(
        os.path.join(output_folder, f"image_{number}_median.png"),
        median_filter
    )
```

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```
cv2.imwrite(
    os.path.join(output_folder, f"image_{number}_median.png"),
    median_filter
)

cv2.imwrite(
    os.path.join(output_folder, f"image_{number}_bilateral.png"),
    bilateral_filter
)

cv2.imwrite(
    os.path.join(output_folder, f"image_{number}_laplacian.png"),
    laplacian_sharp
)

cv2.imwrite(
    os.path.join(output_folder, f"image_{number}_unsharp.png"),
    unsharp
)

cv2.imwrite(
    os.path.join(output_folder, f"image_{number}_highboost.png"),
    high_boost
)

plt.figure(figsize=(16, 12))

plt.subplot(3, 3, 1)
plt.imshow(original_display, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(3, 3, 2)
plt.imshow(mean_filter, cmap="gray")
plt.title("Mean Filter")
plt.axis("off")

plt.subplot(3, 3, 3)
plt.imshow(gaussian_filter, cmap="gray")
plt.title("Gaussian Filter")
plt.axis("off")
```

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```
    plt.subplot(3, 3, 4)
    plt.imshow(median_filter, cmap="gray")
    plt.title("Median Filter")
    plt.axis("off")

    plt.subplot(3, 3, 5)
    plt.imshow(bilateral_filter, cmap="gray")
    plt.title("Bilateral Filter")
    plt.axis("off")

    plt.subplot(3, 3, 6)
    plt.imshow(laplacian_sharp, cmap="gray")
    plt.title("Laplacian Sharpening")
    plt.axis("off")

    plt.subplot(3, 3, 7)
    plt.imshow(unsharp, cmap="gray")
    plt.title("Unsharp Masking")
    plt.axis("off")

    plt.subplot(3, 3, 8)
    plt.imshow(high_boost, cmap="gray")
    plt.title("High Boost Sharpening")
    plt.axis("off")

    plt.suptitle(image_names[i], fontsize=18)
    plt.tight_layout()

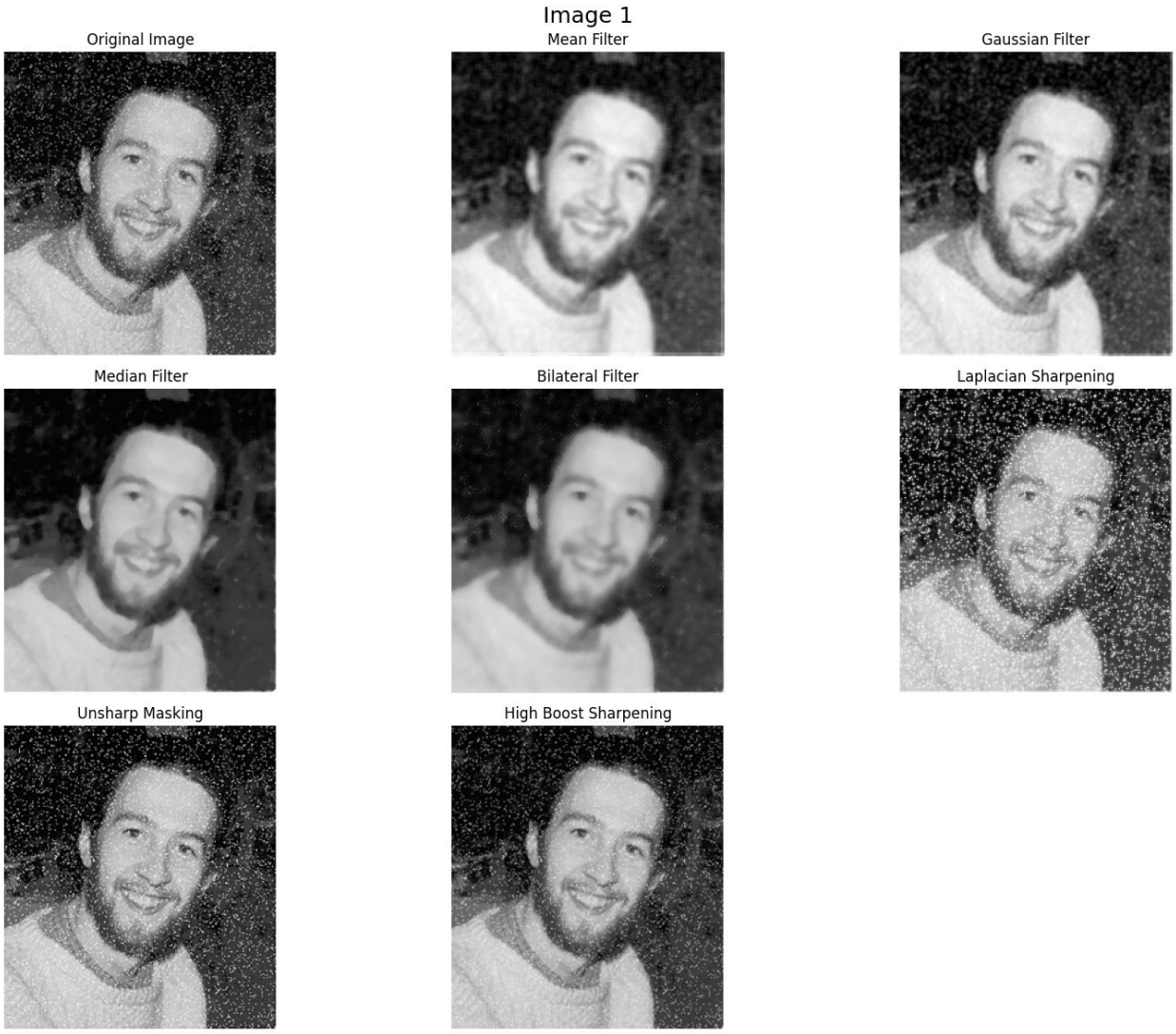
    comparison_file = os.path.join(
        output_folder,
        f"image_{number}_comparison.png"
    )

    plt.savefig(
        comparison_file,
        dpi=300,
        bbox_inches="tight"
    )

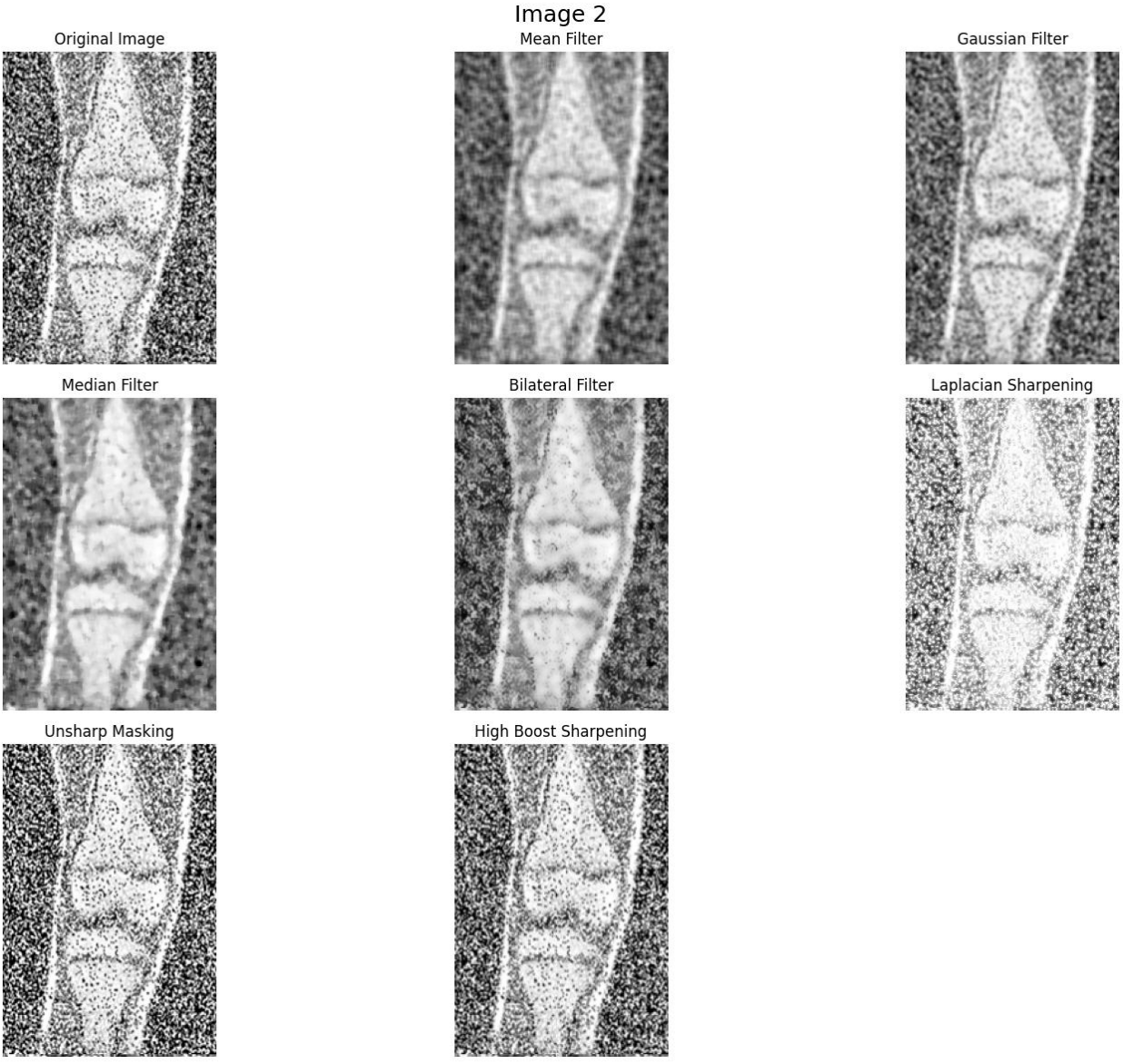
    plt.show()
✓ 10.2s
```

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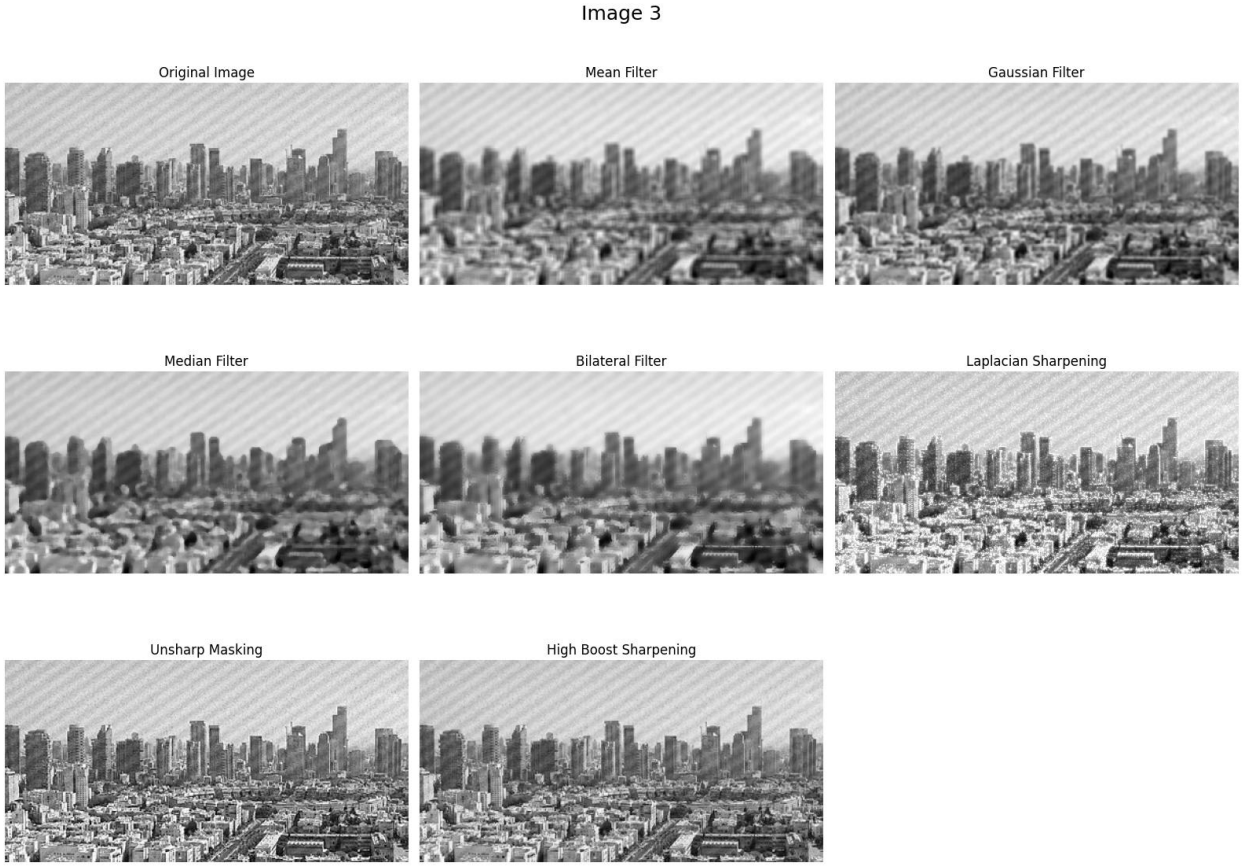
Output: -



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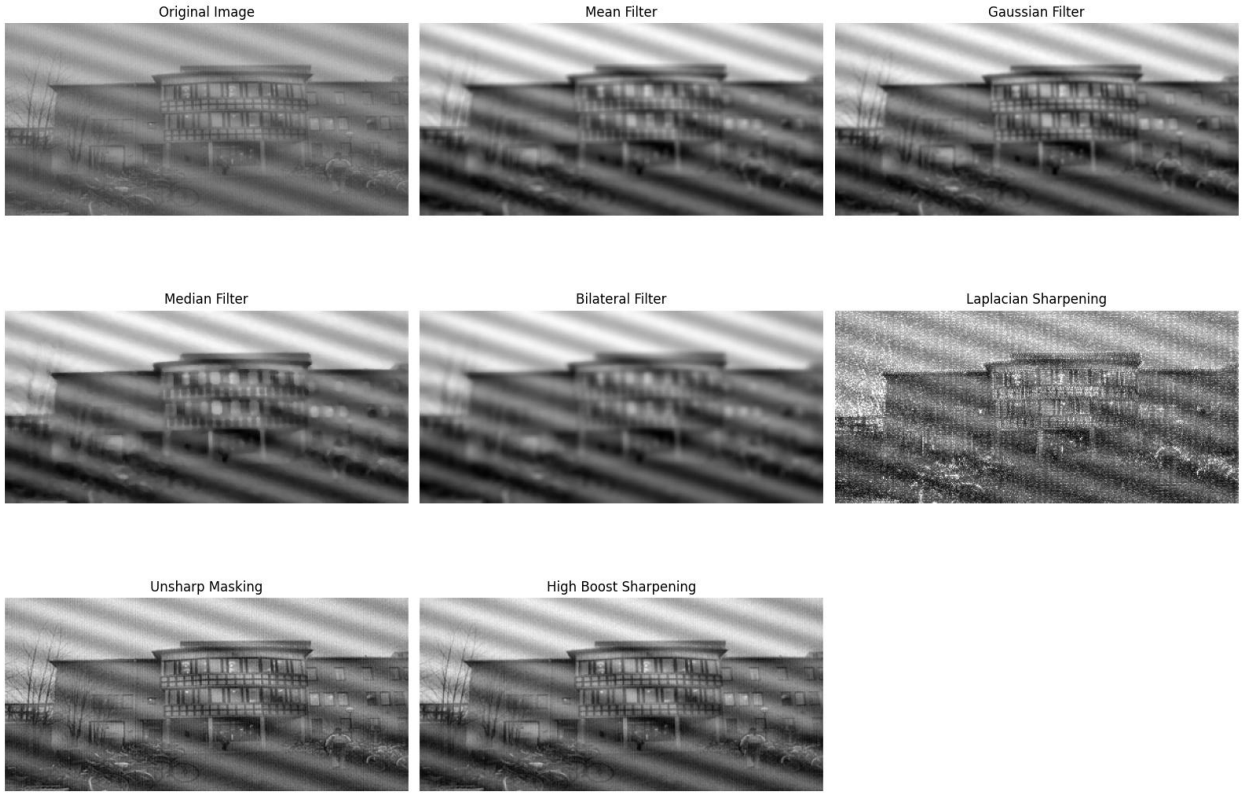


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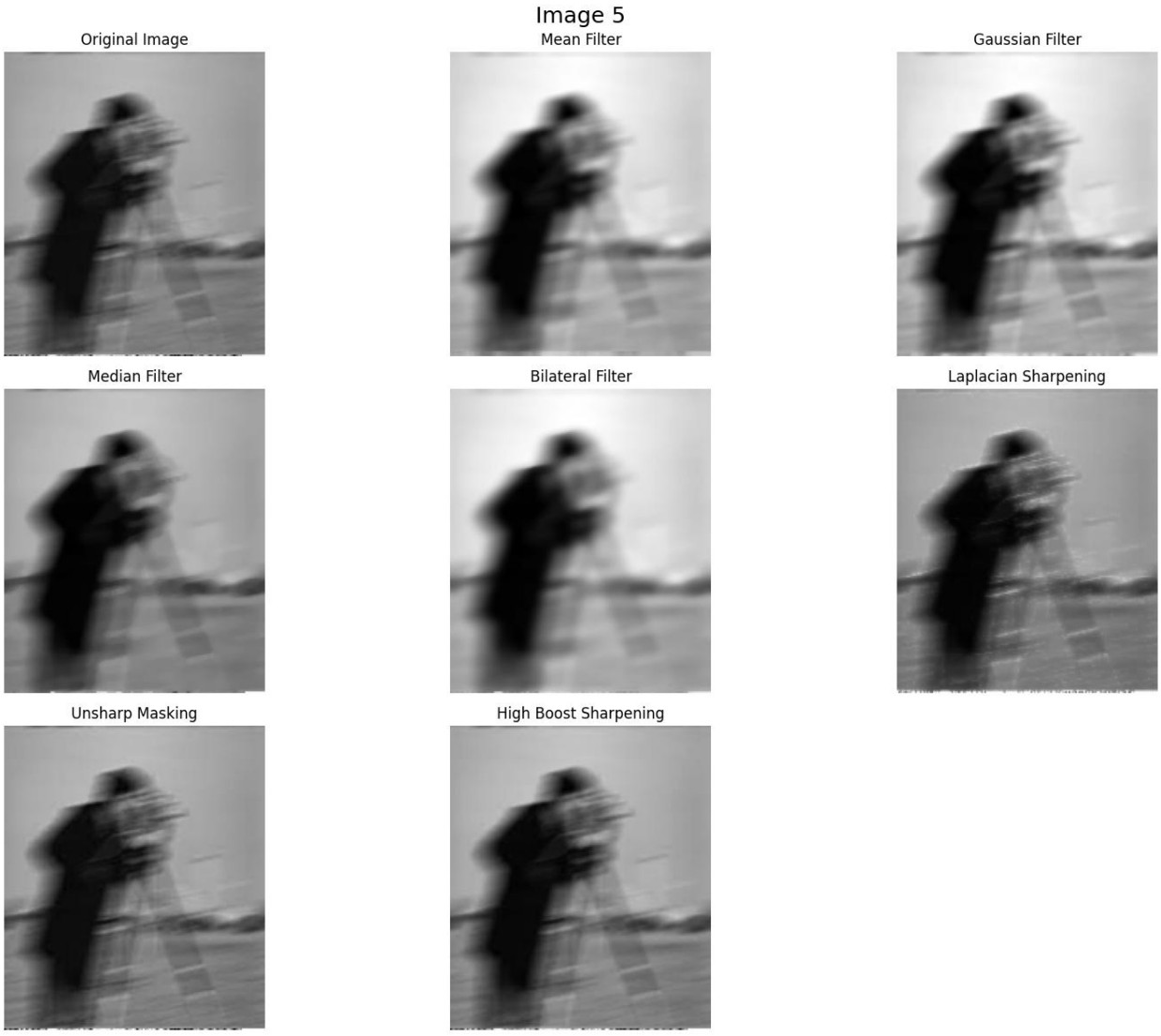


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Image 4



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Observation:

- Mean Filter: Image becomes smooth but slightly blurry.
- Gaussian Filter: Noise is reduced and the image looks smoother.
- Median Filter: Noise is removed and edges are better preserved.
- Bilateral Filter: Noise is reduced while edges remain clear.
- Laplacian Sharpening: Edges become sharper, but noise may also increase.
- Unsharp Masking: Fine details and edges become clearer.
- High Boost Sharpening: Image becomes sharper with stronger edge enhancement.

Justification:

Smoothing filters are used to reduce noise and blur unwanted details. Sharpening filters are used to make edges and fine details clearer. Therefore, different filters are selected according to the required image quality.

Conclusion:

The experiment shows that smoothing filters reduce noise, while sharpening filters improve edges and details. Median and Bilateral filters give good noise reduction, while Unsharp and High Boost give better sharpness.